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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 13: Sides and Angles of a Triangle

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - all four theorems with full proofs in the book's own steps, worked examples, the solved exercise, and the Review Exercise MCQs with a checked answer key.

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> **How to use these notes:** This is a theorem unit. Learn each theorem in four parts - statement, figure, Given / To Prove / Construction, and proof - then practise ordering sides and angles. In the board paper a theorem is usually asked exactly as the book states it, so learn the book's wording.

What this unit covers

1. Theorem 13.1 - the longer side has the greater angle opposite to it
2. Theorem 13.2 - the side opposite the greater angle is longer (the converse)
3. Ordering all three sides and angles of a triangle
4. Theorem 13.3 - the Triangle Inequality: any two sides together are longer than the third
5. Theorem 13.4 - from a point outside a line, the perpendicular is the shortest distance
6. Exercise 13.1 solved, Review Exercise MCQs, and a theorem summary

Source book: KPTB Mathematics for Grade IX, Unit 13 (book pages 267-274). The notes follow the book's order, figures and proofs. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. Opposite Sides and Opposite Angles

Every result in this unit is about one matching pair: a side of a triangle and the angle that does **not** touch it. Before any question, name the pairs correctly.

Side	Angle opposite to it	Angles at its ends
BC (opposite A)	Angle A	Angles B and C
CA (opposite B)	Angle B	Angles C and A
AB (opposite C)	Angle C	Angles A and B

Reading rule: the side opposite an angle is the side between the other two vertices. Side BC does not touch A, so it lies opposite angle A. The measure of a side is written m AB, m BC, m CA, and the measure of an angle is written m angle A, and so on.

Two facts the proofs use again and again

- **Isosceles triangle fact:** angles opposite equal sides are equal, and sides opposite equal angles are equal.
- **Exterior angle fact:** an exterior angle of a triangle is greater than either of the two remote (non-adjacent) interior angles. In symbols, the exterior angle at a vertex is greater than each opposite interior angle.
- **Whole is greater than the part:** if angle CAB is split into angle 1 and angle DAB, then m angle CAB is greater than m angle 1.
- **Only one right or obtuse angle:** a triangle can have only one right angle, and only one obtuse angle, because the three angles add to 180 degrees.

The one idea of the whole unit: bigger side faces bigger angle, and bigger angle faces bigger side. Theorems 13.1 and 13.2 prove this matching in the two directions.

> **Exam trap:** These comparisons work **inside one triangle only**. Do not compare a side of one triangle with an angle of a different triangle unless a proof first connects them.

2. Theorem 13.1 - Longer Side, Greater Angle

Theorem 13.1: If two sides of a triangle are unequal in length, the longer side has an angle of greater measure opposite to it.

Given: Triangle ABC with m BC > m AC. (Side BC is the longer of the two sides being compared.)

To Prove: m angle CAB > m angle B. (Angle CAB is opposite BC; angle B is opposite AC.)

Construction: Take a point D on CB such that CD is congruent to CA. Join D to A. In the figure, angle 1 is angle CAD at A, and angle 2 is angle CDA at D.

Proof - statements and reasons

Statements	Reasons
In triangle ADC: CA is congruent to CD, so angle 1 is congruent to angle 2	Construction; angles opposite to congruent sides of triangle ADC
But m angle 2 > m angle B	Angle 2 is an exterior angle of triangle ABD
Therefore m angle 1 > m angle B --- (i)	Because angle 1 = angle 2
Also m angle CAB > m angle 1 --- (ii)	Because m angle CAB = m angle 1 + m angle DAB; the whole is greater than the part
Therefore m angle CAB > m angle B	From (i) and (ii)

> **Why the construction works:** Cutting off a length equal to AC on the longer side BC creates an isosceles triangle ADC. Its base angles (angle 1 and angle 2) are equal, and angle 2 then beats angle B because it is an exterior angle. Angle CAB is even bigger than angle 1, so it certainly beats angle B.

Corollary worth remembering

In a right-angled triangle, the right angle is the greatest angle, so the side opposite to it - the **hypotenuse** - is the **longest side** of the triangle. This follows from the same matching of bigger angle to bigger side.

3. Theorem 13.2 - Greater Angle, Longer Side

Theorem 13.2: If two angles of a triangle are unequal in measure, the side opposite to the greater angle is longer than the side opposite to the smaller angle.

Given: Triangle ABC in which m angle B > m angle A.

To Prove: m CA > m CB. (CA is opposite angle B; CB is opposite angle A.)

Proof - by trichotomy (only three cases are possible)

Statements	Reasons
Three possibilities: (i) m CA < m CB, or (ii) m CA = m CB, or (iii) m CA > m CB	Trichotomy property - for any two lengths exactly one of these holds
If m CA < m CB, then m angle B < m angle A	By Theorem 13.1, the angle opposite the smaller side is smaller
This is contrary to the given m angle B > m angle A. So case (i) is impossible.	Contradiction with Given
If m CA = m CB, then m angle B = m angle A	Base angles of an isosceles triangle
This is also contrary to the given. So case (ii) is impossible.	Contradiction with Given
Hence the only possibility that holds is m CA > m CB	Cases (i) and (ii) are ruled out

> **Method to name in the exam:** Theorem 13.2 is proved by the **indirect method** - assume the other cases, show each one contradicts what is given, and keep the only case left. Notice it uses Theorem 13.1, so 13.1 must be learned first.

Theorems 13.1 and 13.2 together

- Equal sides face equal angles, and equal angles face equal sides.
- Longer side faces greater angle (13.1); greater angle faces longer side (13.2).
- So the **largest angle** is always opposite the **longest side**, and the **smallest angle** is always opposite the **shortest side**.

4. Ordering All the Sides and Angles

Most short questions in this unit ask for an **increasing order**. The method is always the same two steps: order what is given, then write the matching opposite parts in the same order.

Example 1 (book). List the sides of triangle QRS in increasing order. The angles are: at Q = 30 degrees, at S = 50 degrees, at R = 100 degrees.

Solution: As 30 < 50 < 100, the opposite sides follow the same order: **RS < QR < QS**. (RS is opposite the 30-degree angle at Q; QR is opposite the 50-degree angle at S; QS is opposite the 100-degree angle at R.)

Example 2 (book). List the angles of triangle MNP in increasing order. The sides are: MP = 6, PN = 8, MN = 11.

Solution: As 6 < 8 < 11, the opposite angles follow the same order: **m angle N < m angle M < m angle P**. (Angle N is opposite the side of length 6; angle M is opposite the side of length 8; angle P is opposite the side of length 11.)

Worked example - find the third angle first

Example 3. In triangle ABC, m angle A = 90 degrees, m angle B = 53 degrees and m angle C = 37 degrees. Order the sides. Angles: 37 < 53 < 90, so the opposite sides give **AB < AC < BC**. The hypotenuse BC, opposite the right angle, is longest - exactly as the corollary on page 3 says.

Three-step method for the exam

1. If an angle is missing, find it first: the three angles add to 180 degrees.
2. Write the given angles (or sides) in increasing order.
3. Write the opposite sides (or angles) in exactly the same order, and check each pair once against the table on page 2.

> **Common mistake:** Writing the side that touches the angle instead of the side opposite to it. For angle A the opposite side is BC - the one side that does not touch A at all.

5. Theorem 13.3 - The Triangle Inequality

Theorem 13.3 (Triangle Inequality Theorem): The sum of the lengths of any two sides of a triangle is greater than the length of the third side.

Given: A triangle ABC.

To Prove: m AB + m AC > m BC, and likewise m AB + m BC > m AC and m AC + m BC > m AB.

Construction: Bisect angle A. Let the bisector meet BC at D. Angle 1 and angle 2 are the two halves of angle A; angle 3 is at D on the C side and angle 4 is at D on the B side.

Proof - statements and reasons (for AB + AC > BC)

Statements	Reasons
m angle 1 = m angle 2 --- (i)	AD is the bisector of angle A
But m angle 3 > m angle 1 --- (ii), so m angle 3 > m angle 2	Angle 3 is an exterior angle of triangle ABD; then use (i) and (ii)
This implies m AC > m DC --- (iii)	In triangle ADC, side opposite the bigger angle (Theorem 13.2)
Similarly m angle 4 > m angle 1, which implies m AB > m BD --- (vi)	Angle 4 is an exterior angle of triangle ACD; side opposite the bigger angle
Therefore m AB + m AC > m BD + m DC, that is, m AB + m AC > m BC	Adding (iii) and (vi); and m BC = m BD + m DC

Similarly we can prove that m AB + m BC > m AC and m AC + m BC > m AB.

> **Plain meaning:** The straight route along one side is always shorter than the detour along the other two sides - but the two sides together must still be long enough to meet, so their sum must beat the third side.

6. Using the Triangle Inequality

Range rule: If two sides of a triangle are a and b (with a the longer), the third side x must satisfy **a - b < x < a + b** - greater than the difference, less than the sum. Both ends are strict: equality would flatten the triangle into a straight line.

Example 3 (book). A triangle has one side of length 12 and another of length 8. Describe the possible length of the third side.

Let x be the length of the third side. Small values: x + 8 > 12, so x > 4. Large values: 8 + 12 > x, so x < 20.

The length of the third side must be greater than 4 and less than 20.

Can these three lengths form a triangle?

Lengths	Check: two smaller added vs largest	Triangle?
6 in., 10 in., 3 in. (book Ex. 13.1 Q2)	6 + 3 = 9, which is less than 10	**No**
2 cm, 2 cm, 2 cm	2 + 2 = 4, which is greater than 2	Yes
3 m, 4 m, 5 m	3 + 4 = 7, which is greater than 5	Yes
5 cm, 8 cm, 2 cm	5 + 2 = 7, which is less than 8	**No**
3 m, 3 m, 2 m	3 + 2 = 5, which is greater than 3	Yes
1.5 m, 5 m, 3.5 m	1.5 + 3.5 = 5, equal to 5 - not greater	**No**
2.5 cm, 3.5 cm, 4.5 cm	2.5 + 3.5 = 6, which is greater than 4.5	Yes

Short cut: Only one check is needed - add the two smaller lengths and compare with the largest. If that sum is not greater, no other pairing can save it.

Example 4. Two sides of a triangle are 11 and 23 and the third side is x (book Ex. 13.1 Q4). By the range rule: 23 - 11 < x < 23 + 11, so **12 < x < 34**.

7. Theorem 13.4 - The Perpendicular Is Shortest

Theorem 13.4: From a point outside a line, the perpendicular is the shortest distance from the point to the line.

Given: KL is perpendicular to line XY, where K is a point outside XY and L is the foot of the perpendicular on XY. In the figure, angle 1 is the angle at L in triangle KLM (the right angle), and angle 2 is the angle at M.

To Prove: m KL is the shortest of all the distances from K to line XY.

Construction: Take a point M on XY other than L. Join M to K. Triangle KLM is formed.

Proof - statements and reasons

Statements	Reasons
In triangle KLM: m angle 1 = 90 degrees	Given - KL is perpendicular to XY
Therefore m angle 1 > m angle 2	Only one angle can be a right angle in a triangle, so angle 2 is less than 90 degrees
Or m KM > m KL	Side opposite to the greater angle (Theorem 13.2)
Or m KL < m KM	Same inequality, read the other way

Similarly it can be proved that the distance of K from any point taken on XY except L will be greater than KL. Hence the perpendicular KL is the shortest distance of point K from line XY.

What this theorem is used for

- **Distance from a point to a line** always means the perpendicular distance - never a slant distance.
- **Distance between two parallel lines** is the length of any perpendicular segment between them, and it is the same everywhere. This is the key to Exercise 13.1 Q9: if PL and QM are the shortest distances between the parallel lines AB and CD, both are perpendicular to the same lines, so PL is parallel to QM.
- Every **altitude** of a triangle is a perpendicular distance, which is why each altitude is shorter than the sides meeting at its vertex (used in Exercise 13.1 Q10).

8. Exercise 13.1 - Solved

1. **Sides 7 and 12, third side x:** x + 7 > 12 gives x > 5, and 7 + 12 > x gives x < 19. So **5 < x < 19**.
2. **Sticks 6 in., 10 in., 3 in.:** 6 + 3 = 9 < 10, so **no triangle can be formed**.
3. **Base QR of isosceles triangle PQR extended to S; prove m PS > m PQ:** The equal sides give equal base angles, m angle PQR = m angle PRQ, and each base angle is less than 90 degrees. Angle PRQ is an exterior angle of triangle PRS at R, so m angle PSR < m angle PRQ. Also m angle PRS = 180 - m angle PRQ, which is greater than m angle PRQ. Hence in triangle PRS, m angle PRS > m angle PSR, so by Theorem 13.2 the opposite side is longer: m PS > m PR = m PQ.
4. **Sides 11 and 23, third side x: 12 < x < 34** (worked on page 7).
5. **S on side PQ of triangle PQR, S joined to R; prove PQ + QR > PS + SR:** In triangle SQR, SQ + QR > SR (Theorem 13.3). Add PS to both sides: PS + SQ + QR > PS + SR. But PS + SQ = PQ, so **PQ + QR > PS + SR**.
6. **Two angles 45 degrees and 70 degrees:** the third angle is 180 - 45 - 70 = 65 degrees. (i) The side opposite 70 degrees is greater than the side opposite 45 degrees. (ii) The shortest side is opposite the smallest angle, 45 degrees. (iii) The longest side is opposite the greatest angle, 70 degrees.
7. **Triangle RST with RS > ST and m angle S = 60 degrees:** RS > ST means angle T > angle R (Theorem 13.1). Since angle R + angle T = 120 degrees and angle T is the larger of the two, angle T > 60 degrees > angle R. (i) Smallest side: ST, opposite the smallest angle R. (ii) Longest side: RS, opposite the largest angle T.
8. **Sets that can be sides:** (a) 2, 2, 2 - yes. (b) 3, 4, 5 - yes. (c) 5, 8, 2 - no. (d) 3, 3, 2 - yes. (e) 1.5, 5, 3.5 - no. (f) 2.5, 3.5, 4.5 - yes. (Full checking table on page 7.)
9. **AB parallel to CD; PL and QM the shortest distances between them; prove PL parallel to QM:** Shortest distance means perpendicular (Theorem 13.4), so PL and QM are both perpendicular to CD. Two lines perpendicular to the same line are parallel to each other, hence **PL is parallel to QM**.
10. **Perimeter greater than the sum of the altitudes:** Each altitude is the perpendicular distance to a side, so by Theorem 13.4 it is shorter than each slant side from the same vertex. Adding the three such inequalities pairs each altitude with the sides of the triangle and gives: sum of altitudes < perimeter of the triangle.

9. Review Exercise 13 - MCQs and Summary

Multiple-choice questions (book Review Exercise 13)

1. An exterior angle measures 120 degrees and one remote interior angle is 40 degrees. The second remote interior angle is: (a) 40 (b) 80 (c) 70 (d) 120
2. In triangle ABC, angle A = 90, angle B = 53, angle C = 37 degrees. The sides satisfy: (a) AB < BC < CA (b) AC < BC < AB (c) AB < AC < BC (d) BC < AC < AB
3. P lies outside line AB. PR will be the shortest distance if angle PRA is: (a) 180 (b) 45 (c) 100 (d) 90 degrees
4. K lies outside line XY and KB is perpendicular to XY. The shortest distance is: (a) m KD (b) m KC (c) m KA (d) m KB
5. Two sides are 10 and 14. The third side can be: (a) 2 (b) 4 (c) 22 (d) 24
6. In triangle ABC, angle A = 50 and angle B = 30 degrees. Correct is: (a) m BC > m AB (b) m AB > m CA (c) m BC < m CA (d) m AB < m CA
7. Which represents the sides of a triangle? (a) 3, 4, 5 (b) 3, 4, 7 (c) 3, 4, 8 (d) 3, 4, 1
8. In triangle KLM, angles K = 45, L = 55, M = 80 degrees. The longest side is: (a) KL (b) LM (c) KM (d) none of these

Answer key: 1(b) - 120 - 40 = 80. 2(c). 3(d). 4(d). 5(c) - the third side must lie between 4 and 24, and 4 itself is not allowed. 6(b) - angle C = 100 degrees is largest, so AB is longest. 7(a). 8(a) - KL is opposite the largest angle, 80 degrees.

Review short questions - answers

Two sides 8 in. and 12 in.: the third side lies between **4 in. and 20 in.** Can sides be 7 in., 5 in., 12 in.? **No** - 7 + 5 = 12, which is not greater than 12.

Theorem summary - learn these four lines

Theorem	Statement
13.1	Longer side has the greater angle opposite to it.
13.2	Side opposite the greater angle is longer than the side opposite the smaller angle.
13.3	Sum of any two sides is greater than the third side.
13.4	From a point outside a line, the perpendicular is the shortest distance to the line.

> **Final tip:** For ordering questions, match the angle order with the opposite-side order. For "can it be a triangle?", add the two smaller sides - they must beat the largest side, strictly.